Self Driven Synchronous Rectification in Resonant Topologies: Forward ZVS-MRC, Forward ZCS-QRC and LCC-PRC

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ABSTRACT

Improvements in IC's integration scale demand lower supply voltages, being 3.3V a present-day trend, and 1.5V a probable standard in the next years. This trend is completely opposite to high efficiency and high power density, the most important specifications in On-Board converters for distributed power systems. Losses in the rectifiers of the output stage are the main constraint to obtain high efficiency in low output voltage power supplies. These losses may be minimized applying synchronous rectification techniques. Self-driven synchronous rectification in resonant topologies is an important chance to overcome these problems. Three resonant topologies (LCC-PRC, Forward ZCS-QRC and Forward ZVS-MRC) with self-driven synchronous rectification are analyzed in this paper.

INTRODUCTION

Low output voltage DC/DC converters have a very important constraint: high efficiency is a very difficult task to achieve. This fact is due to the voltage drop in the rectifier diodes of the output stage. Efficiency decreases drastically when this voltage is of the same order of magnitude than the output voltage.

Let us assume a 0.6V diode voltage drop (V_d) and a 3.3V output voltage (V_o) converter. In this case, and taking into account only losses in the rectifier stage (P_d), efficiency is limited to 84% (see eq. 1), since output power (P_o) is not much higher than P_d.

$$\eta = \frac{P_o}{P_d + P_o} = \frac{I_o \cdot V_o}{I_o \cdot V_d + I_o \cdot V_o} = \frac{V_o}{V_d + V_o} = \frac{3.3}{0.6 + 3.3} = 84\% \quad (1)$$

If we consider now all the losses in the converter, overall efficiency would hardly be over 75%. Such low efficiency produces serious inconveniences: temperature increases, heat sinks are needed, size increases and hence power density and reliability decrease.

These features are very important for On Board DC/DC converters in Distributed Power Architectures. The scope of this paper is telecom applications. The converter is connected to a 48V battery (38V $\leq V_{in} \leq$ 58V), and supplies 20A output current. Present-day output voltage is 5V, but it will be 3.3V in the next future.

Since our main goal is to improve efficiency, it is convenient to analyze power losses distribution in dc-dc converters, in order to determine how converter losses may be minimized.

Figure 1 shows losses distribution in both PWM and Resonant converters. It can be seen that this distribution is different in each case, due to the different behaviour the primary side of the converter. However, since output stage behaviour of is very similar in both cases, losses in rectifiers are almost the same, being around 50%. When output voltage is 3.3V, these losses become around 60% of the overall losses.

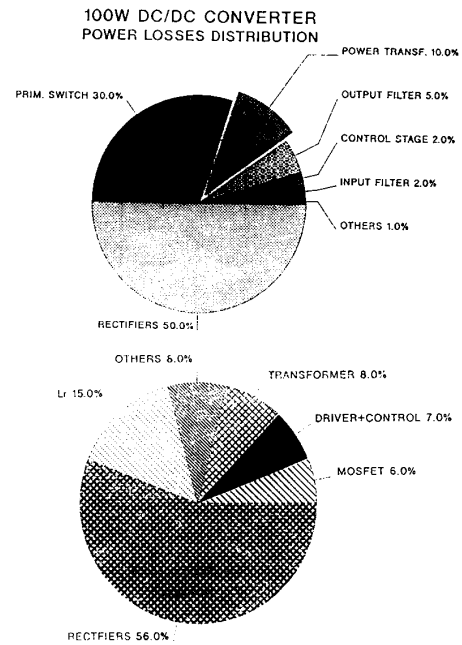

Fig. 1. Losses distribution in both PWM (a) and Resonant Converters (b) (5V Output Voltage)

It is clear that the higher improvements will be achieved optimizing the rectifier stage. Thus, the only way to keep efficiency at reasonable levels is to substitute diodes by synchronous rectifiers, featuring lower voltage drop ([1]-[5]). At high switching frequencies the most suitable device to be used as synchronous rectifier to be used as synchronous rectifier is the power MOSFET. This way, the almost constant diode voltage drop, is substituted by a resistance type (R_{DSon}) voltage drop. This resistance can be minimized paralleling devices, but the problem when several MOSFETs are connected in parallel is that the value of parasitic capacitances increase.

This effect is an inconvenience when square-wave external driving is used, because driving losses are $C_{iss}V^2_{GS}f_s$ type and are very affected by the value of MOSFET parasitic capacitances. This driving strategy is not recommended at high switching frequencies in resonant converters.

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Fortunately MOSFET synchronous rectifiers may be driven in a non-dissipative fashion in resonant topologies. However, not all resonant topologies are suitable for self-driven synchronous rectification [6]. Three resonant topologies have been built for telecom applications, and are compared in this paper.

SELF DRIVEN SYNCHRONOUS RECTIFICATION IMPLEMENTATION IN RESONANT CONVERTERS

The major advantage of resonant converters from the point of view of synchronous rectification is that driving losses can be avoided.

The main two requirements that the resonant topology must fulfil are that the synchronous rectifier parasitic capacitances must be included in the power stage and there must also be a resonant capacitance able to switch on and off the synchronous rectifiers. These two requirements are equivalent to have a sinusoidal type voltage waveform in the secondary winding of the transformer and a constant current source behaviour of the output filter (LC filter).

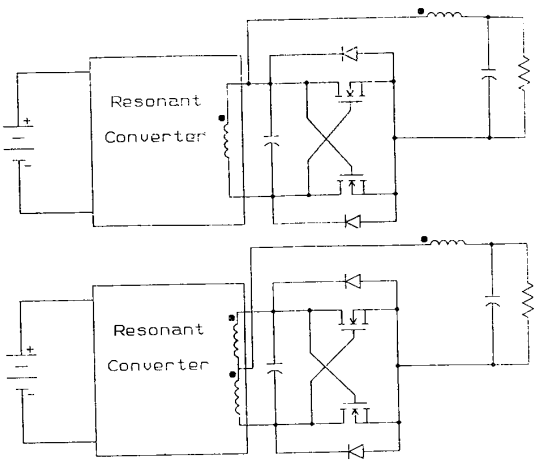

Fig. 2. Self-driven Synchronous Rectification Implementation in resonant topologies

The implementation of self-driven synchronous rectification in these topologies is shown in figure 2. The structure of the synchronous rectifier stage is always the same. The voltage waveform in the secondary winding of the transformer is almost sinusoidal, and it is the driving waveform of both synchronous rectifiers. This stage has been tested with a low frequency sinusoidal driving waveform (line waveform, 50Hz) in order to analyze its behaviour without the typical undesirable oscillations that appear at high switching frequency.

In this structure, voltage in the gate of one MOSFET (V_{GS1}) is also voltage in the drain of the other MOSFET (V_{DS2}), since the gate of each MOSFET is connected to the drain of the other. Figure 3 shows both waveforms at 5V/div (V_{GS}) and 100mV/div (V_{SD}). We can see that each MOSFET is driven by a sinusoidal waveform. When this voltage is lower than MOSFET threshold voltage, it is off and the output current flows through the schottky diode. The voltage drop during this time is approximately 400mV.

When voltage in the gate reaches threshold voltage MOSFET conducts, and voltage drop is, in this case (10A output current) around 60mV. When voltage in the gate is again lower than threshold voltage MOSFET turns off and schottky diode conducts. The voltage drop is again 400mV. During the other semicycle the behaviour is exactly the same, since both, driving waveform and synchronous rectifier stage, are symmetrical.

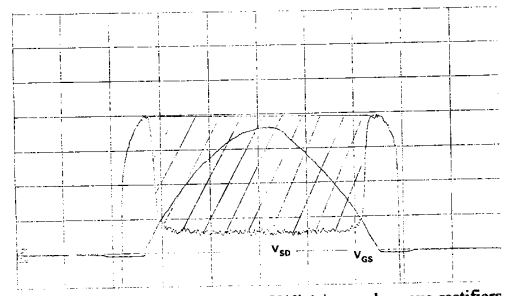

Fig. 3. V_{GS} (5V/div) and V_{SD} (100mV/div) in synchronous rectifiers at line (50Hz) frequency (2ms/div)

Power saved with self-driven synchronous rectification is proportional to the difference between diode and MOSFET voltage drop during the time that the MOSFET conducts. To obtain the maximum benefit from this technique, MOSFET should feature a very low on-resistance (R_{DSon}) in order to reduce the voltage drop, and also a very low threshold voltage (V_{TH}), in order to increase the time that MOSFET conducts.

We have not taken into account losses in the gate yet, because we have assumed that MOSFETs are driven in a self-driving fashion by the secondary winding voltage waveform, and hence the typical driving losses disappear.

However, MOSFET parasitic capacitances are connected in parallel with a resonant capacitance, and then part of the resonant current flows through MOSFETs gates. Figure 4 shows the equivalent circuit of any resonant topology with self-driven SR. In it we can see that part of the resonant current flows through the ESR of the discrete resonant capacitor and other part flows through MOSFETs gate resistance (R_g). If R_g is higher than ESR there are additional losses that should be taken into account.

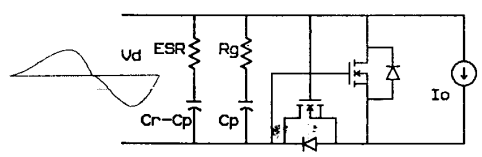

Fig. 4. Output stage equivalent circuit

Gate Resistance is the most important MOSFET parameter, because it limits the reduction of the on-resistance (R_{DSon}) due to the increase of parasitic capacitances.

Summarizing, MOSFET SR specifications are:

- Low On Resistance (↓ R_{DSon})
- Low Gate Resistance (↓ R_g)
- Low Threshold Voltage (↓ V_{TH})
- Parasitic Capacitances are not restricted if R_g is low enough.

COMPARATIVE ANALYSIS OF RESONANT TOPOLOGIES WITH SELF DRIVEN SYNCHRONOUS RECTIFICATION

The objective of this paper is to select, among the resonant topologies suitable for self-driven synchronous rectification those that are specially adequate for telecom applications. The main goal is to obtain high power density on-board converters for distributed power systems. Input voltage is a 48V bus ($38 \leq V_{in} \leq 58V$) and the output current ranges from 10A to 20A. The converters have been analyzed for both 3.3V and 5V output voltage.

The three resonant topologies selected have been Forward ZVS-MRC (figure 5), Forward ZCS-QRC (figure 6) and LCC-PRC (figure 7). Forward ZVS-MRC and LCC-PRC topologies have been optimized for 5V output voltage, and later they have been tested at 3.3V.

In the case of Forward ZCS-QRC, two specific converters have been designed for each output voltage, following the design procedure described in [7]. The values of the resonant parameters are shown in the figures.

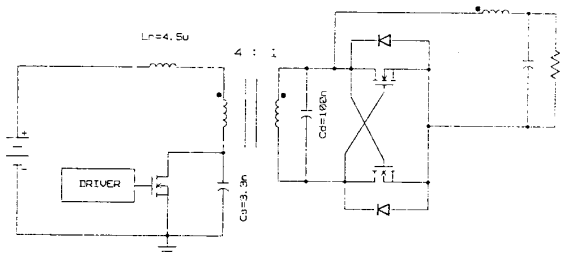

Fig. 5. Forward ZVS-MRC with self-driven Synchronous Rectification

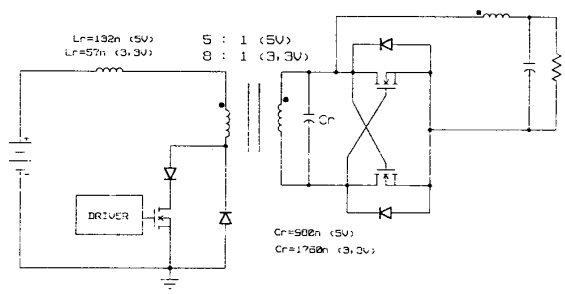

Fig. 6. Forward ZCS-QRC with self-driven Synchronous Rectification

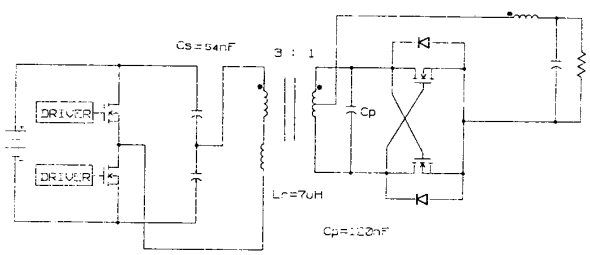

Fig. 7. LCC-PRC with self-driven Synchronous Rectification

These three topologies have been selected due to their simplicity and their suitability for self-driven synchronous rectification. The output stage is the same (resonant capacitor in the secondary winding of the transformer rectifier stage and output filter), with the only difference that the LCC-PRC is center-tapped, whereas Forward ZVS-MRC and Forward ZCS-QRC are single ended.

Then, the only difference from the point of view of synchronous rectification is the driving waveform in the secondary winding of the transformer. In the case of Forward ZVS-MRC the optimization of the power stage is not strongly dependent of the use or not of synchronous rectification. However, in the Forward ZCS-QRC it is convenient to design the power stage, and above all, the transformer, following the design procedure described in [7], in order to increase the slope of the driving waveform at the crossovers.

In the case of LCC-PRC care must be taken when designing the power stage in order to avoid operation points that feature voltage waveforms at the secondary winding of the transformer like the one shown in figure 8. It is because in such waveform MOSFET/diode conduction ratio is very low.

Another difference among these topologies from the point of view of synchronous rectification is switching frequency. As it is shown in next figures, switching frequency in Forward ZVS-MRC is higher than in the other topologies, what produces more parasitic

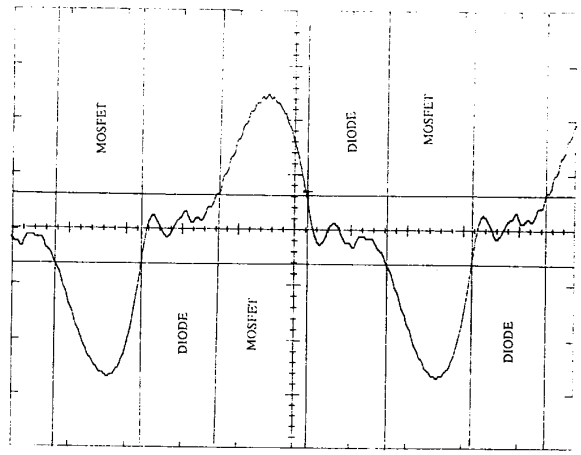

Fig. 8. Driving waveform in an improper design of a LCC-PRC

oscillations in the synchronous rectifiers.

Regarding the primary side, Forward ZVS-MRC and LCC-PRC have two resonant elements, whereas Forward ZCS-QRC has only one. Regarding the switches, the two Forward topologies have only one but with higher stresses (200V maximum) than the two primary MOSFETs of LCC-PRC (60V maximum)

Finally, Forward ZCS-QRC has a diode connected in series with the primary MOSFET, what produces additional losses.

Figures 9, 11, 13 show the voltage waveform at the secondary winding of the transformer, that is the driving waveform. It should be noted that there is a small oscillation at the turn on of the MOSFETs, in both positive and negative cycles. These oscillations do not appear in figure 3 because frequency is much lower (50Hz) and because the synchronous rectifier stage is supplied by the line voltage, and not by a resonant capacitor.

Figures 10, 12, 14 show the voltage drop (V_{SD}) in the MOSFETs. The behaviour is the same that in figure 3, but with oscillations due to the high switching frequencies. These oscillations may be reduced using pockaging techniques, that have not been used in these converters.

All these figures show that there is an amount of power saved, what increases efficiency. MOSFETs used as synchronous rectifiers (SMP60N03) feature low on-resistance (R_{DSon} = 10mΩ) and low threshold voltage (≃3V), but gate resistance was not optimized (R_g ≃ 1.5Ω).

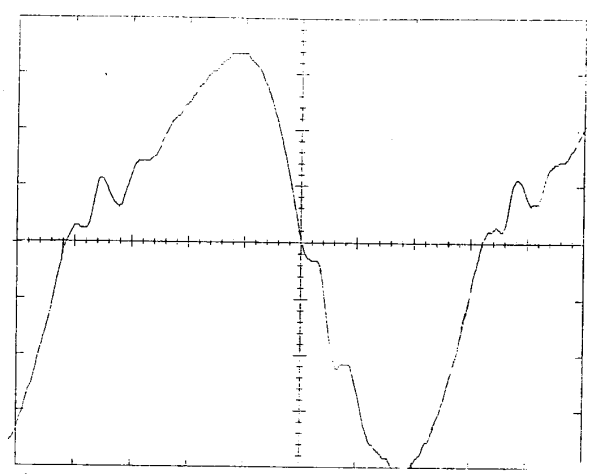

Fig. 9. Driving waveform (secondary winding) at full load (48V, 20A) in the Forward ZVS-MRC

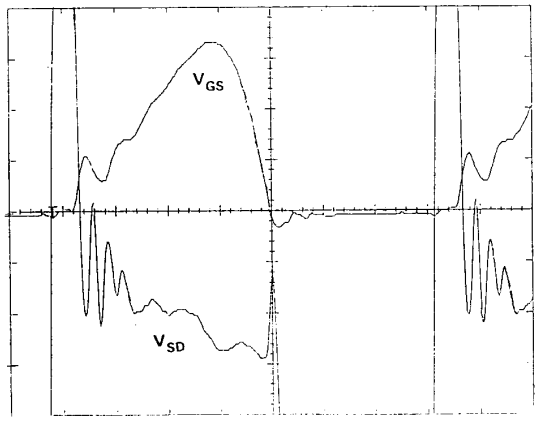

Fig. 10. V_{GS} (5V/div) and V_{SD} (100mV/div) in Forward ZVS-MRC

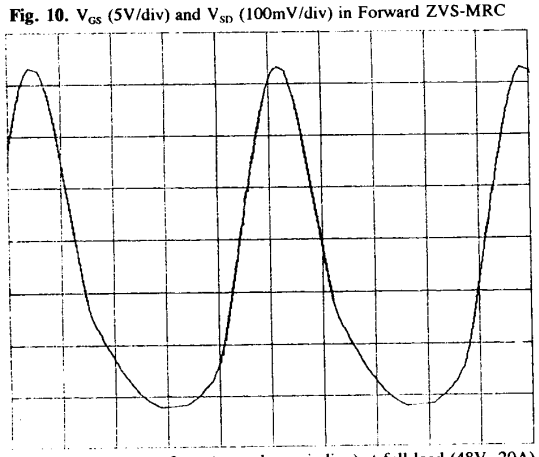

Fig. 11. Driving waveform (secondary winding) at full load (48V, 20A) in the Forward ZCS-QRC

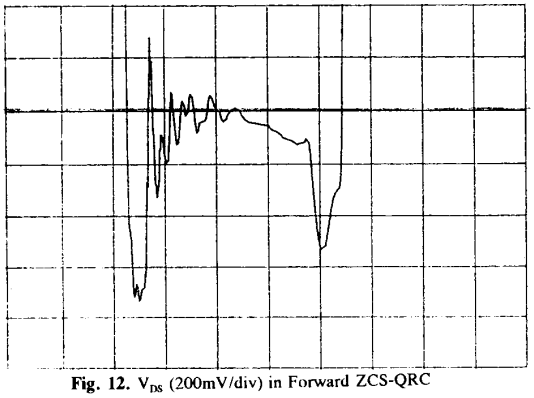

Fig. 12. V_{DS} (200mV/div) in Forward ZCS-QRC

Converters were tested using three MOSFETs connected in parallel in each branch. If gate resistance were lower, more MOSFETs could be connected in parallel, and better results would be achieved.

Figures 15, 16, 17 show switching frequency and efficiency measured in the three resonant topologies in the case of 5V output voltage.

Although the specifications only required variations in the output current from 10A to 20A, Forward ZVS-MRC and LCC-PRC have the ability to operate from no load to 20A and measures in this range are presented. It should be noted that switching frequency variation is not too high for none of this topologies.

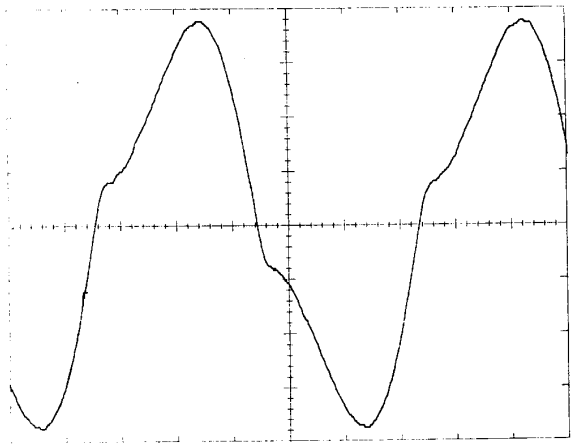

Fig. 13. Driving waveform (secondary winding) at full load (48V, 20A) in the LCC-PRC

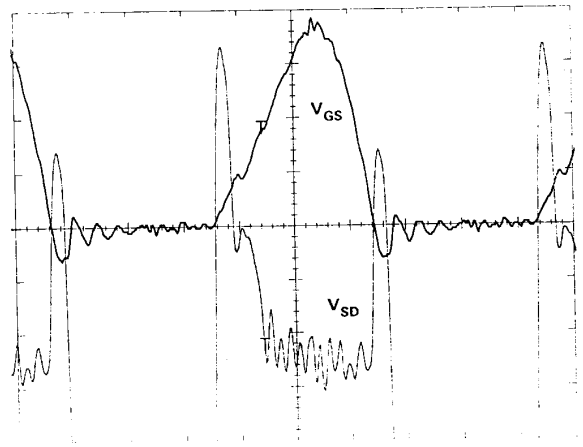

Fig. 14. V_{GS} (5V/div) and V_{SD} (100mV/div) in LCC-PRC

Regarding the efficiency, the maximum value was obtained in both Forward converters at full load (87.8 and 87.6%). In the MRC it was achieved with low input voltage (38V), whereas in the QRC it was achieved at high input voltage (58V). When the output current decreases, efficiency decreases very much in the case of the QRC, whereas it is over 80% even at 5A output current in the case of MRC.

Efficiency is also very good in the case of LCC-PRC being the maximum value at half-load (10A, 86%). It is over 80% from 5A to 15A, but is lower at full load. The main advantage of this converter is that the resonant current is very affected by the load, decreasing when it decreases, and keeping efficiency at high levels even at light loads.

Forward ZVS-MRC and LCC-PRC were also tested at 3.3V output voltage without modifying resonant parameters, and so the design was no optimazed. Switching frequency and efficiency measured are showm in figures 18 and 20. Forward ZCS-QRC was optimized for 3.3V output voltage, and the values of the resonant parameters are shown in figure 6. Switching frequency and efficiency measured in this converter are shown in fig 19.

Switching frequency variation ranges are very similar to those obtained with 5V output voltage. Maximum efficiency obtained was 84% at 15A in the case of MRC, and 82.8% in the QRC at full load (20A). Maximum efficiency obtained in LCC-PRC was 78% at light load (5A). Efficiency decreases when the load increases because the design is not optimum and the circulating energy increases very much with the load.

The same effects described at 5V happens at 3.3V in both Forward topologies. Efficiency is high in a wider range in the case of the MRC, being over 80% even at 7A output current.

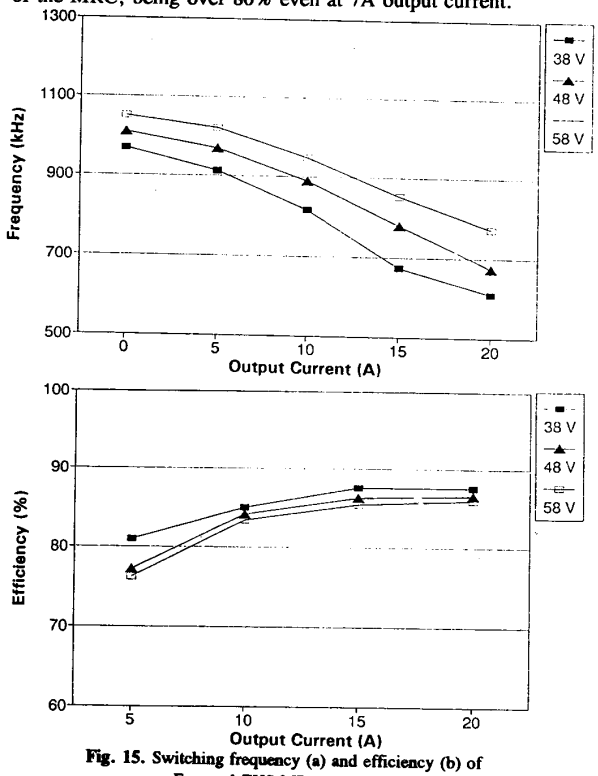

Fig. 15. Switching frequency (a) and efficiency (b) of Forward ZVS-MRC (V_o=5V)

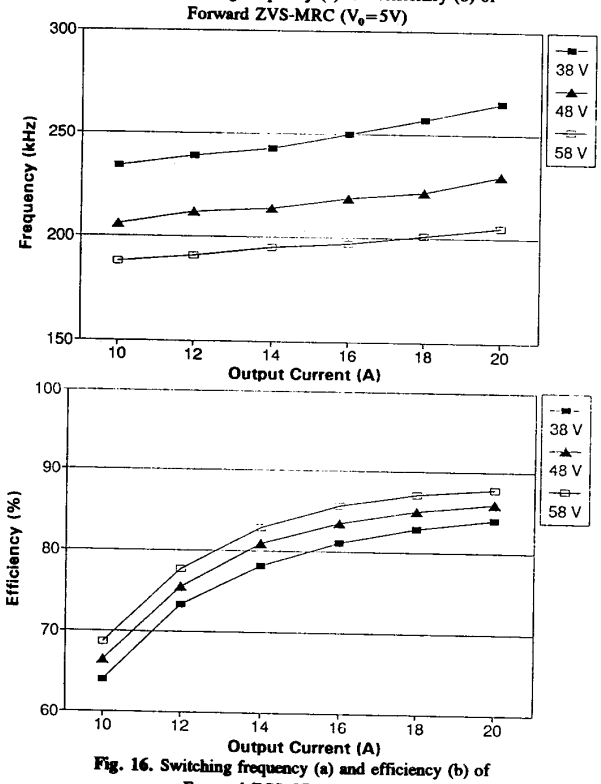

Fig. 16. Switching frequency (a) and efficiency (b) of Forward ZCS-QRC (V_o=5V)

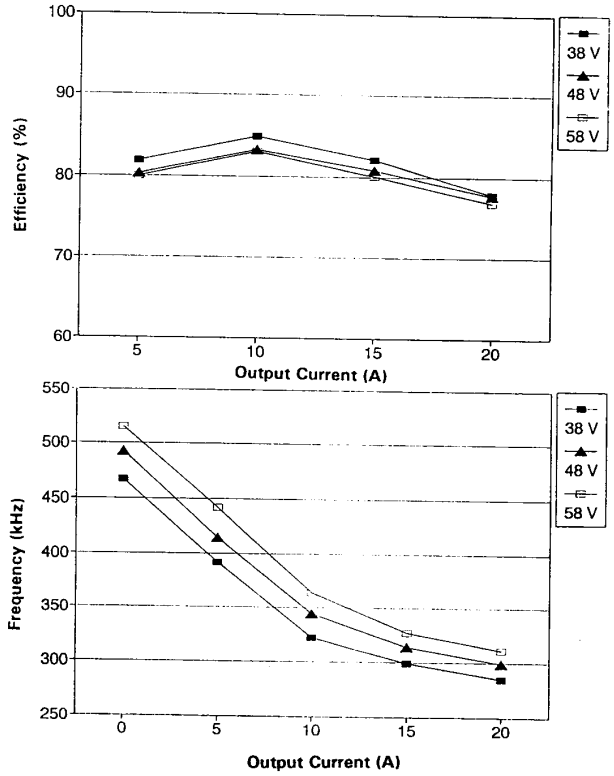

Fig. 17. Switching frequency (a) and efficiency (b) of LCC-PRC (V_o=5V)

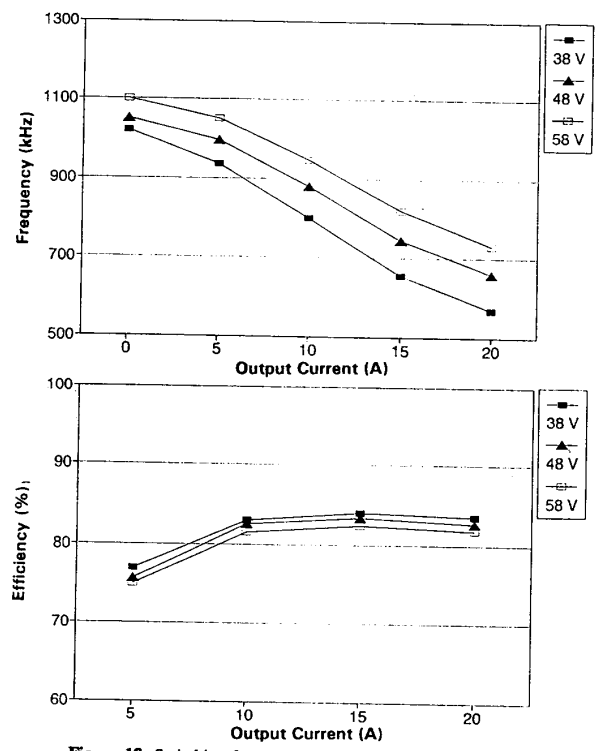

Figure 18. Switching frequency (a) and efficiency (b) of Forward ZVS-MRC (V_o = 3.3V)

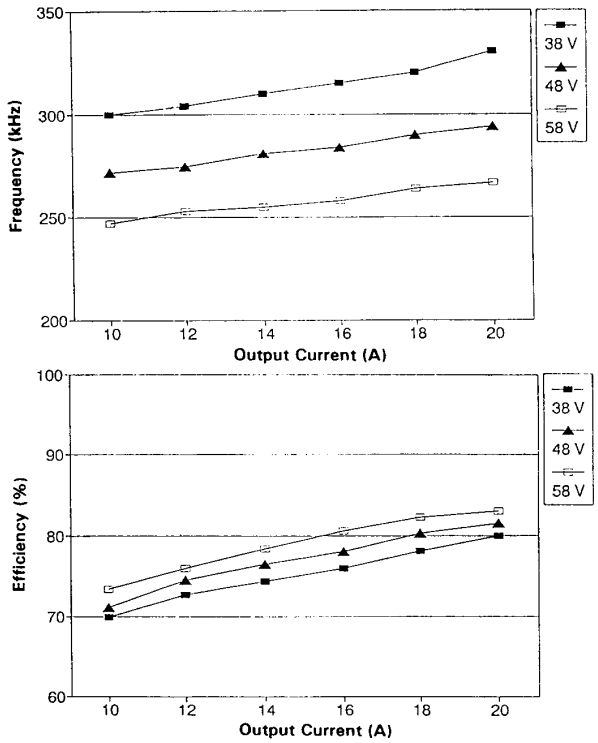

Fig. 19. Switching frequency (a) and efficiency (b) of Forward ZVS-MRC (V_o 3.3V)

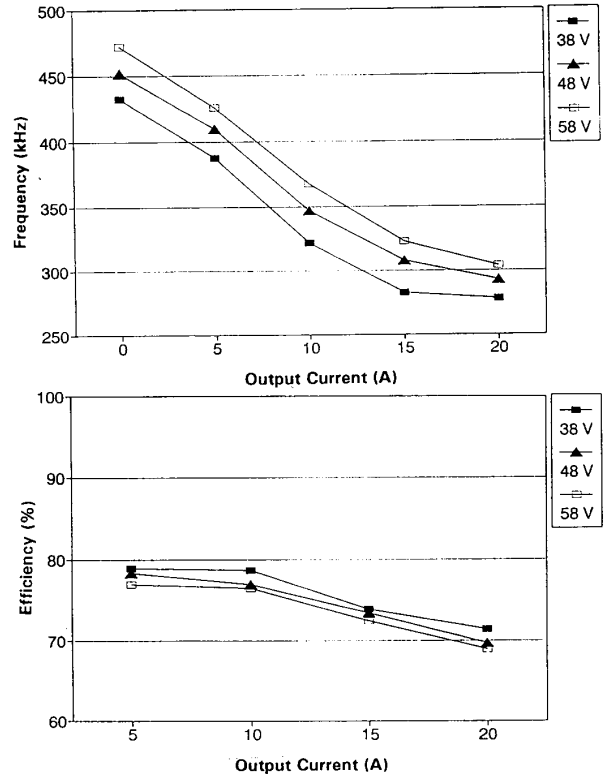

Fig. 20. Switching frequency (a) and efficiency (b) of LCC-PRC (V_o = 3.3V)

CONCLUSIONS

Self-driven Synchronous Rectification in resonant topologies is a very good solution to obtain high efficiency in low output voltage DC-DC converters. A great improvement has been obtained using commercial MOSFETs as synchronous rectifiers, and better results are expected with specific custom made MOSFET devices with lower gate resistance.

Three resonant topologies Forward ZVS-MRC, Forward ZCS-QRC and LCC-PRC with self-driven synchronous rectification have been compared from the point of view of telecom applications, taking into account efficiency, simplicity and switching frequency variation range.

Once these topologies hace been analyzed, the main conclusion is that it is possible to obtain a very high efficiency even with low output voltage (3.3V) in any of these topologies with the use of synchronous rectification. However, when there is a high variation range in load and input voltage the best performance is the one shown by the Forward ZVS-MRC. This topology presents the additional advantage of being able to operate at very high switching frequency.

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